anaesthesia and dissociative anaesthetics. Autacoids. Antimicrobials and principles of chemotherapy in microbial infections. Use of hormones in therapeutics-chemotherapy of parasitic infections. Drug and economic concerns in the Edible tissues of animals- chemotherapy of Neoplastic diseases. Toxicity due to insecticides, plants, metals, non-metals, zootoxins and mycotoxins.

1.6 Veterinary Hygiene with reference to water, air and habitation—Assessment of pollution of water, air and soil—Importance of climate in animal health-effect of environment on animal function and performance-relationship between industrialization and animal agriculture-animal housing requirements for specific categories of domestic animals viz. pregnant cows and sows, milking cows, broiler birds- stress, strain and productivity in relation to animal habitation.

2. Animal Diseases :

2.1 Etiology, epidemiology pathogenesis, symptoms, postmortem lesions, diagnosis, and control of infectious diseases of cattle, sheep and goat, horses, pigs and poultry.

2.2 Etiology, epidemiology, symptoms, diagnosis, treatment of production diseases of cattle, horse, pig and poultry.

2.3 Deficiency diseases of domestic animals and birds.

2.4 Diagnosis and treatment of non-specific conditions like impaction, Bloat, Diarrhoea, Indigestion, dehydration, stroke, poisoning.

2.5 Diagnosis and treatment of neurological disorders.

2.6 Principles and methods of immunization of animals against specific diseases-herd immunity-disease free zones-'zero' disease concept- chemoprophylaxis.

2.7 Anaesthesia—local, regional and general-preanesthetic medication. Symptoms and surgical interference in fractures and dislocation. Hernia, choking abomasal displacement-Caesarian operations. Rumenotomy-Castrations.

2.8 Disease investigation techniques—Materials for laboratory investigation—Establishment of Animal Health Centers-Disease free zone.

3. Veterinary Public Health :

3.1 Zoonoses—Classification, definition, role of animals and birds in prevalence and transmission of zoonotic diseases-occupational zoonotic diseases.

3.2 Epidemiology—Principle, definition of epidemiological terms, application of epidemiological measures in the study of diseases and disease control. Epidemiological features of air, water and food borne infections. OIE regulations, WTO, sanitary and phytosanitary measures.

3.3 Veterinary Jurisprudence—Rules and Regulations for improvement of animal quality and prevention of animal diseases—State and central

rules for prevention of animal and animal product borne diseases—S P C A-Veterolegal cases Certificates Materials and Methods of collection of samples for veterolegal investigation.

4. Milk and Milk Products Technology :

4.1 Market Milk—Quality, testing and grading of raw milk. Processing, packaging, storing, distribution, marketing, defects and their control. Preparation of the following milks: Pasteurized, standardized, toned, double toned, sterilized, homogenized, reconstituted, recombined and flavoured milks. Preparation of cultured milks, cultures and their management, yoghurt, Dahi, Lassi and Srikhand. Preparation of flavoured and sterilized milks. Legal standards. Sanitation requirement for clean and safe milk and for the milk plant equipment.

4.2 Milk Products Technology—Selection of raw materials, processing, storing , distributing and marketing milk products such as Cream, Butter, Ghee, Khoa, Channa, Cheese, condensed, evaporated, dried milk and baby food, Ice cream and Kulfi; by-products, whey products, butter milk, lactose and casein. Testing, grading, judging milk products—BIS and Agmark specifications, legal standards, quality control and nutritive properties. Packaging, processing and operational control. Costing of dairy products.

5. Meat Hygiene and Technology :

5.1 Meat Hygiene :

5.1.1 Ante mortem care and management of food animals, stunning, slaughter and dressing operations; abattoir requirements and designs; Meat inspection procedures and judgment of carcass meat cuts-grading of carcass meat cuts-duties and functions of Veterinarians in wholesome meat production.

5.1.2 Hygienic methods of handling production of meat-Spoilage of meat and control measures-Post-slaughter physicochemical changes in meat and factors that influence them-Quality improvement methods–Adulteration of meat and detection-Regulatory provisions in Meat trade and Industry.

5.2 Meat Technology :

5.2.1 Physical and chemical characteristics of meat—Meat emulsions : Methods of preservation of meat—Curing, canning, irradiation, packaging of meat and meat products, processing and formulations.

5.3 By-products—Slaughter house by-products and their utilization-Edible and inedible by products- Social and economic implications of proper utilization of slaughter house by-products—Organ products for food and pharmaceuticals.

5.4 Poultry Products Technology—Chemical composition and nutritive value of poultry meat, pre-slaughter care and management. Slaughtering techniques, inspection, preservation of poultry meat and products. Legal and BIS standards.